PROJECT REPORT

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

A ServiceNow workflow automation project

GROUP NO.	**GROUP – 4**
PROJECT AREA	ServiceNow – Flow Designer

Prepared as an academic project report based on the Group 4 project title provided.

1. ABSTRACT

Service desks receive many IT support tickets every day. Each ticket may describe an issue such as a login problem, network failure, software error, hardware fault, or access request. When tickets are classified manually, the process can take time and may lead to inconsistent assignment. This project, **Auto Ticket Classification Using Flow Designer**, describes a workflow that can automatically identify the category and priority of an incoming ticket and route it to the appropriate support team.

The proposed solution uses ServiceNow Flow Designer to automate actions when a ticket is created or updated. The flow checks selected ticket details, such as short description, description, category, impact, and urgency. Based on configured conditions, it sets the ticket category and assignment group. It can also add a work note so that the action is clear to the support team. The aim is to reduce repetitive manual work, improve consistency, and help tickets reach the correct team sooner.

Project overview

Item	Description
Project title	Auto Ticket Classification Using Flow Designer
Platform	ServiceNow
Main tool	Flow Designer
Input	New or updated incident/ticket details
Output	Category, priority and assignment group values

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2. INTRODUCTION

An IT service desk is a central point where users report technical problems and request assistance. Every reported issue is recorded as a ticket or incident. The support team must understand the issue, classify it, decide its priority, and assign it to the correct team. Correct classification is important because it helps the organization manage workload and respond to users efficiently.

2.1 Existing system

In a manual ticket-handling process, a service desk agent reads each ticket and selects its category, priority, and assignment group. This method depends on the agent's experience and the quality of the information entered by the user. During busy periods, manual classification can create delays. Similar issues may also be classified differently by different agents.

2.2 Problem statement

The main problem addressed by this project is the repetitive effort required to classify and route incoming tickets. Incorrect or incomplete classification can cause a ticket to reach the wrong team, increasing the time needed to resolve the issue. A consistent automated workflow is therefore useful for common and clearly identifiable ticket types.

2.3 Proposed system

The proposed system uses a ServiceNow Flow Designer flow. When a qualifying ticket is created, the flow evaluates configured conditions and updates ticket fields. For example, a ticket containing a clearly identified network issue can be assigned to the network support group. A login or account-access issue can be routed to the service desk or access support group, depending on the organization's configuration.

2.4 Scope

- Automate classification for selected, common ticket types.
- Set category and assignment group using administrator-defined conditions.
- Use impact and urgency values to support priority assignment where required.
- Keep an exception path for unclear or unmatched tickets so they can be reviewed manually.

Important: Flow Designer follows the conditions configured by the administrator. It does not automatically understand every sentence like a human. Reliable results depend on clear rules, good ticket data, and regular testing.

3. OBJECTIVES AND REQUIREMENTS

3.1 Objectives

- Reduce the time spent manually classifying routine tickets.
- Apply consistent category and assignment rules.
- Route tickets to the appropriate support group.
- Use impact and urgency to support consistent priority setting.
- Maintain an exception route for tickets that do not match known rules.
- Make the workflow easier to monitor and maintain through Flow Designer.

3.2 Functional requirements

ID	Requirement
FR-01	The system should start the flow when a new incident is created.
FR-02	The flow should read the available ticket fields.
FR-03	The flow should check configured conditions for known issue types.
FR-04	The flow should update the category and assignment group.
FR-05	The flow should use an exception path when no rule matches.
FR-06	The flow should record a work note where configured.

3.3 Non-functional requirements

- **Usability:** The flow should be understandable to administrators.
- **Reliability:** Conditions should be tested to avoid incorrect routing.
- **Security:** Only authorized users should configure or administer the flow.
- **Maintainability:** Rules should be easy to review and update.

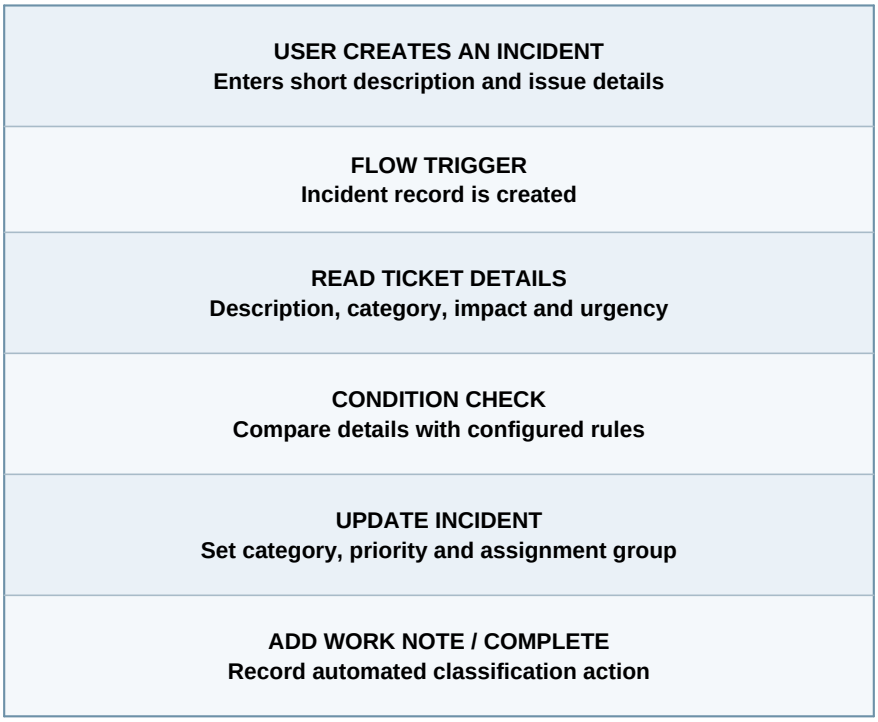
3.4 Basic requirements

Software: ServiceNow instance with access to Incident records and Flow Designer; a supported web browser; administrator or flow-designer permissions as required by the instance. **Data:** Sample incidents with short description, description, category, impact, urgency, and assignment group fields.

4. SYSTEM DESIGN AND WORKFLOW

The system follows an event-driven workflow. The trigger starts the flow when a new incident is inserted. The flow then evaluates the ticket details, chooses a matching rule, and updates the record. If no rule matches, the incident remains available for manual classification.

4.1 Workflow diagram



↓ Each step passes its record data to the next step

4.2 Example classification rules

Ticket details (example)	Category	Assignment group
Cannot connect to office Wi-Fi	Network	Network Support
Unable to sign in / password issue	Account / Access	Service Desk
Printer is not responding	Hardware	Hardware Support
Application is showing an error	Software	Application Support

These are illustrative rules. Actual category values and group names must match the ServiceNow instance used for the project.

5. IMPLEMENTATION USING FLOW DESIGNER

The following steps describe how the proposed automation can be configured in ServiceNow. Exact field names, available actions, and permissions may differ depending on the ServiceNow release and the instance configuration.

5.1 Create the flow

- Open the ServiceNow application and navigate to **Flow Designer**.
- Select **New** and create a flow with a clear name, such as **Auto Ticket Classification**.
- Choose the Incident table or the relevant ticket table as the trigger source.
- Set the trigger to run when a record is created. Add conditions if only selected incidents should be processed.

5.2 Add decision conditions

Add an If / Else If / Else decision structure. Each branch checks for a known issue pattern or a selected field value. For example, a condition may check whether the short description contains a configured keyword such as “Wi-Fi” or “printer”. Keyword rules should be specific enough to reduce false matches.

5.3 Update the incident

- For a matching branch, add an Update Record action for the triggering incident.
- Set the Category field to the appropriate configured value.
- Set the Assignment group to the relevant support group.
- If required, set priority using the organization's impact-and-urgency matrix.
- Add a work note such as “Ticket classified automatically by Flow Designer.”

5.4 Configure the default branch

The Else branch should not force an uncertain classification. It can leave the category and assignment group unchanged, assign the ticket to a triage group, or add a note requesting manual review. The selected approach should follow the organization's service desk process.

5.5 Save, test and activate

Save the flow, use the available test or debugging options with sample records, review the execution details, and correct any condition or field-mapping errors. Activate the flow only after the test cases have been reviewed. Monitor the flow after activation and update rules when ticket patterns change.

Implementation caution

Avoid creating multiple flows that repeatedly update the same incident without clear conditions. This can cause repeated executions or unexpected field changes. Use trigger conditions and, where appropriate, a field or status check to prevent unnecessary reprocessing.

6. TESTING AND EXPECTED RESULTS

Testing should confirm that each rule selects the correct category and assignment group, and that unmatched tickets are not classified incorrectly. The following sample cases can be used during demonstration. They are test scenarios, not a claim that the flow has already been executed in a live ServiceNow instance.

Test	Sample ticket	Expected behaviour
TC-01	Cannot connect to Wi-Fi	Category: Network; assign to Network Support.
TC-02	Password reset required	Category: Account / Access; assign to Service Desk.
TC-03	Office printer not working	Category: Hardware; assign to Hardware Support.
TC-04	Business app shows an error	Category: Software; assign to Application Support.
TC-05	Need help with something	No rule match; route for manual review.

6.1 Expected benefits

- **Faster handling:** Routine tickets can be classified without waiting for an agent to perform every field update.
- **Consistency:** The same configured rule is applied to similar tickets.
- **Better routing:** Tickets can reach the intended support group earlier.
- **Reduced repetitive work:** Service desk agents can spend more time on investigation and resolution.
- **Traceability:** Flow execution details and work notes can help administrators review automated actions.

6.2 Limitations

Keyword-based rules may fail when users use unexpected words, spelling mistakes, or descriptions containing several issues. A ticket may also need more than one support team. Therefore, automated classification should support—not completely replace—human review. Rules must be tested with realistic examples and monitored for incorrect assignments.

7. CONCLUSION

Auto Ticket Classification Using Flow Designer provides a structured way to automate the classification and routing of common IT support tickets in ServiceNow. A flow can start when an incident is created, examine selected fields, apply configured conditions, and update the category and assignment group. An exception path helps ensure that unclear tickets remain available for manual review.

The project demonstrates how low-code workflow automation can reduce repetitive service desk activities and improve consistency. The quality of the outcome depends on the accuracy of the rules, the ticket information entered by users, and the correct configuration of assignment groups and priority values. Testing and monitoring are necessary before the flow is used in a production environment.

7.1 Future enhancements

- Use more detailed decision rules and maintain a configurable keyword list.
- Add duplicate-ticket checks and identify tickets that may describe the same issue.
- Send notifications to the assigned support group after classification.
- Track classification accuracy and the number of tickets routed automatically.
- Explore AI-based classification only where the ServiceNow instance has the required licensed capability, data quality, and governance.

REFERENCES

- ServiceNow Product Documentation – Flow Designer (official ServiceNow documentation).
- ServiceNow Product Documentation – Incident Management and incident record fields.
- ServiceNow Product Documentation – Flow triggers, conditions, actions, and Update Record operations.

Note for project demonstration: Before submission, replace the illustrative rule names with the exact categories and assignment groups available in your ServiceNow instance. Add screenshots of your own Flow Designer configuration and test executions if your faculty requires implementation evidence.

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